Fault Masking and Diagnosis in Reversible Circuits

Masoud Zamani
Electrical and Computer Eng.
Northeastern University
Boston, USA
Email: mzamani@ece.neu.edu

Navid Farazmand
Electrical and Computer Eng.
Northeastern University
Boston, USA
Email: nfarazma@ece.neu.edu

Mehdi B. Tahoori
Faculty of Informatics
Karlsruhe Institute of Technology
Karlsruhe, Germany
Email: mehdi.tahoori@kit.edu

***Abstract*—Reversible logic is a promising design methodology, particularly in the scope of quantum computing, for extremely low power consumption by elimination of power dissipation due to information loss. Anticipated high fault rates for future technologies raise demand for fault tolerance in reversible logic. In this paper we propose fault masking techniques (to prevent error propagation) for reversible logic. We present different implementations of reversible majority voters. Using voter insertion techniques and taking advantage of available non-functional outputs, we are able to provide diagnosis information with adjustable resolution for reversible circuits. Such diagnosability has important applications in manufacturing testing as well as online repair. In contrast to previous work this voter is robust against single point of failure. We target missing and repeated gate fault models which are specific to reversible logic.**

***Keywords*-Reversible logic; Majority voter; Fault masking;**

I. Introduction

Reversible logic has received considerable attention in quantum computing [1], [2]. Quantum logic gates operation must be based on elementary unitary matrix operations which are reversible. In addition to importance of reversible logic in quantum computing, it is considered as a promising design paradigm for ultra low-power computation in future emerging technologies, because of the possibility of nearly energy-free computation in this logic [1]. It should be noted that implementing reversible logic in current VLSI technologies does not lead to reduction in power dissipation since currently the major source of power dissipation is not due to information loss. Quantum computing as a physical realization of reversible logic motivates further research in this domain.

An n-input, n-output gate is called reversible if it realizes a reversible function by a one to one mapping between input and output combinations. In other words, a reversible logic function is a boolean function which maps each input combination to a unique output combination. It means that, in a reversible circuit data is bijectively transformed without losing original information [3]. Equality in the number of inputs and outputs is a necessary condition for a reversible function [4], [5]. A reversible circuit consists of reversible gates without fan-out branches and feedbacks.

High failure rate is predicted to be one of the major challenges in any future technology. Hence, online error detection and correction/masking become an important parts of the design of reversible circuits.

Full controllability and observability is one of the important properties of reversible circuits. This property simplifies test generation for off-line testing of reversible circuits [6]. It has been shown that, for any reversible circuit, there is a complete single stuck-at fault test set. The same complete test set for single stuck-at fault also detects all multiple stuck-at faults in the circuit [7]. Online error detection techniques have been proposed for reversible circuits. These techniques can be categorized as parity-generating (e.g. [8]), parity-preserving (e.g. [9]), and dual rail online error detection methods [10]. All these methods target stuck-at fault model, however, it is shown this fault model is not appropriate for reversible circuits [11]. Therefore, researchers have proposed specific fault models including *Missing Gate Fault* (MGF) and *Repeated Gate Fault* (RGF) models for reversible circuits. In these models fault is characterized as elimination (replication) of gates in the circuit [12]. These fault models are based on physical characteristics of reversible circuits. Therefore, reliable design techniques must explicitly address these fault types.

In this paper, we present fault masking techniques for reversible circuits and provide two reversible majority voters with different error masking capabilities. These reversible voters have different diagnosis and error detection capabilities with different area overheads. Non-functional outputs of these voters (garbage outputs) are used in providing fault diagnosis information while fault is masked on functional outputs. A set of reversible modules are also proposed to provide adjustable diagnosis resolutions. Beside the fault masking capability to ensure the reliability of the circuit, the diagnosis capability of the voters reduces complexity of manufacturing test as well as runtime and in-field repairs. A major advantage of these voters with respect to previous work is their robustness against single point of failure. In contrast to previous work, these voters are efficient in dealing with and tolerating MGF and RGF faults.

The organization of the rest of the paper is as follows. Section II briefly introduces preliminaries and related work. The proposed majority voters are presented in Section III. Fault diagnosis scheme is discussed in Section IV. Experimental results and comparison of proposed schemes are presented in Section V. Finally, Section VI concludes the paper.

1530-1877/11 $26.00 © 2011 IEEE
DOI 10.1109/ETS.2011.19

IEEE computer society

II. Preliminaries and Related Work

A. Preliminaries

In a reversible circuit there is a one to one relation between inputs and outputs. A set of universal reversible gates which are known as Fredkin Gate, Feynman Gate, and Toffoli Gate are introduced to implement all reversible circuits [13]. As seen in Fig. 1, inputs of these gates are divided to two groups: target inputs (T), and control inputs (C). Outputs are also categorized as garbage outputs (G), and function outputs (F). All logic gates (except inverters) require garbage outputs as well as control inputs to satisfy one to one mapping between inputs and outputs. For the Toffoli gate (To) each G_i has the same value as corresponding C_i. For this gate; $F = (\prod C_i) \oplus T$. A Toffoli gate which has only one control line is known as Feynman Gate (Fe). In a Fredkin Gate (Fr) for each i, $G_i = C_i$. If all C_is are 1, then $F_1 = T_2$, and $F_2 = T_1$, otherwise $F_1 = T_1$, and $F_2 = T_2$.

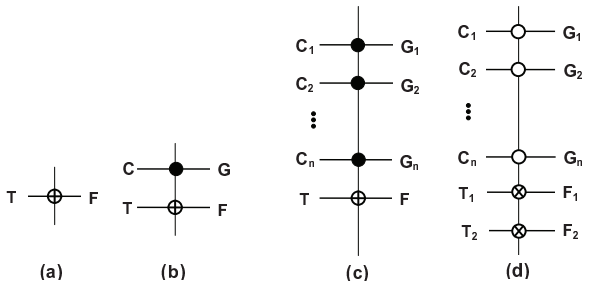

Figure 1. Universal reversible gates: (a) Not (b) Feynman (Fe) (c) Toffoli (To) (d) Fredkin (Fr)

B. Related Work

Reversibility in the circuit significantly simplifies off-line testing compared to irreversible circuits. It has been shown that there is a test set which can detect all single/multiple stuck-at faults in a reversible circuit [6]. Using parity in online testing of reversible circuits has widely been investigated (e.g. [8]). Parity preserving approach is also used for online testable design of reversible circuits [9].

New logical fault models based on the concepts of missing and repeated gate operations at the logical level have been introduced in [11], [12]. These fault models are more realistic for reversible logic implementations compared to traditional VLSI fault models, e.g. single stuck-at fault [14]. These models are called *Single Missing Gate Fault* (SMGF) model, *Multiple Missing Gate Fault* (MMGF), *Partial Missing Gate Fault* (PMGF), and *Repeated Gate Fault* (RGF). SMGF is a fault model which assumes a reversible gate has been removed (completely) from the circuit. MMGF assumes more than a gate (but all are consecutive to each other) have removed (completely) from the circuit. RGF assume a gate is replaced by two consecutive copies of that gate. PMGF assumes a part of a gate has been removed [11], [12].

Failure analysis of *Triplicated Modular Redundancy* (TMR) in conventional irreversible logic has been explored in [15]. Various distributions of TMR and voters in a circuit have been also presented. Logic synthesis on a set of reversible gates called *Majority Based Reversible Logic Gate* (MBRLG) has been discussed in [16]. An MBRLG is a reversible gate with odd number of inputs such that at least one of the gate outputs is a majority Boolean function of all inputs. A voting technique for reversible logic has been introduced based on majority multiplexing [17] . A reversible majority gate called *MAJ* was proposed [17]. In this module one of the output bits is the majority of the input bits. This gate suffers from single point of failure problem. To resolve this problem, authors have used this gate similar to the Von-Neumann multiplexing method, which results in area overhead more than $6X$ (due to six copies of the module and its reverse as well as fan-out branch gates).

III. Reversible Majority Voters

In this section two different implementations for a majority voter with triplicated voted outputs are proposed. These modules provide fault masking and fault diagnosis in the scope of TMR. The schematic of such voter is shown in Fig. 2. It consists of three data inputs (I_1, I_2, and I_3), and two control lines in the input of the module (C_1, and C_2). It generates three data outputs (O_1, O_2, and O_3), and two garbage outputs (D_1, and D_2). Voting is done on data inputs, and results are produced on data outputs. The objective of majority voter implementation is producing all 0 or all 1 on three data outputs. In other words, two permutations out of eight possible permutations of these outputs are used in the implementation. On the other hand, reversibility impose one to one mapping between input and output permutations. Therefore, at least two garbage outputs are required to preserve reversibility. In this work the two garbage outputs which are unavoidable due to reversibility requirements will be exploited to perform fault diagnosis. Since in a reversible circuit fan-out on wires is not possible, the proposed voters provide three copies of the voted output to enable efficient implementation of distributed TMR architectures. In addition, the triplicated outputs can be implemented in such a way that eliminates the problem of single point of failure in the voter.

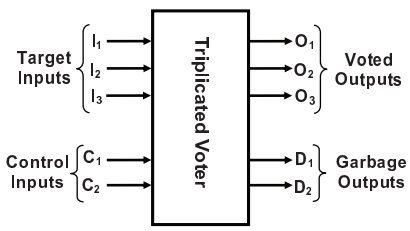

Figure 2. Proposed reversible majority voter, with triplicated voted output

A. Area-Efficient Implementation

Minimal Triplicated Voter (MTV) is an implementation of the described voter which is shown in Fig. 3. The main objective in MTV implementation is using minimum number of stages. This implementation consists of 4 stages (i.e. 4 reversible gates). The relation between I_i ($1 \leq i \leq 3$)

and O_i $(1 \leq i \leq 3)$ is shown in Table I. This table also represents the effect of faulty input on D_1 and D_2.

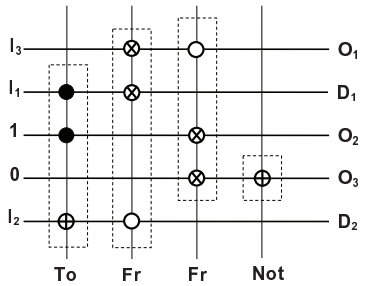

Figure 3. Minimal Triplicated Voter (MTV)

Table I
INPUT-OUTPUT RELATION OF THE MTV

Inputs	Outputs	
I_3 I_2 I_1	O_3 O_2 O_1	D_2 D_1
0 0 0	0 0 0	0 0
0 0 1	0 0 0	1 0
0 1 0	0 0 0	1 1
1 0 0	0 0 0	0 0
1 1 1	1 1 1	0 1
1 1 0	1 1 1	1 0
1 0 1	1 1 1	1 1
0 1 1	1 1 1	0 1

Analysis of MTV: Our fault model includes missing-gate and repeated-gate faults. Bit-flip on signal lines (including the inputs of the voters) is also considered in our analysis which could be the effect of missing or repeated gate in previous stage. A single fault results in three types of errors as follows:

1) *Maskable errors*: those who can be masked or result in error on garbage lines which do not affect functionality of the circuit.
2) *Recoverable errors*: if a fault results in bit-flip on one of the three copies of voted output, then it can be recovered by the voter at successor TMR stage. Therefore, the effect of this fault can be recovered if no other fault(s) occur in that successor voter.
3) *Unrecoverable errors*: any fault which changes more than one copy of the voted outputs, results in an error which is neither maskable nor recoverable.

Any single fault on inputs of the MTV is masked by this module (Maskable error). Furthermore, a single fault inside the MTV results in Maskable or Recoverable errors. However, it is not robust against missing gate and repeated gate faults which occur in the voter while there is a fault on the input lines of voter. In addition to this, the diagnosis information of this voter is not sufficient to indicate faulty input. In order to increase the robustness of the voter as well as increasing the diagnosis information of the voter another implementation of the voter is proposed.

B. Robust Implementation

The proposed *Robust Triplicated Voter (RTV)* is shown in Fig. 4. The input-output relation of RTV is shown in Table II. In MTV implementation, the output is calculated based on the value of inputs. Then three copies of the output are generated. In contrast to MTV implementation, in RTV implementation each copy of voted value is produced independently, directly from the inputs. Since RTV produces three independent copies of the majority input bits, it reduces number of faulty cases which result in Unrecoverable error. Also, most of the single faults are maskable in RTV rather than being recoverable. This module provides full diagnosis for single faults. Note that, these capabilities are achieved at the cost of increased area and delay (in terms of circuit depth).

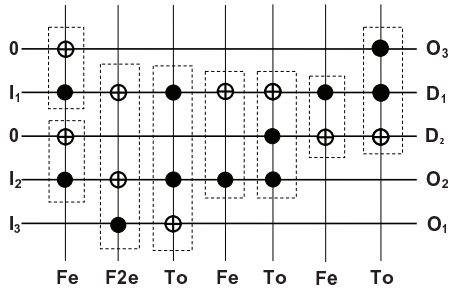

Figure 4. Robust Triplicated Voter (RTV)

Table II
INPUT-OUTPUT RELATION OF THE RTV

Inputs	Outputs	
I_3 I_2 I_1	O_3 O_2 O_1	D_2 D_1
0 0 0	0 0 0	0 0
0 0 1	0 0 0	0 1
0 1 0	0 0 0	1 0
1 0 0	0 0 0	1 1
1 1 1	1 1 1	0 0
1 1 0	1 1 1	0 1
1 0 1	1 1 1	1 0
0 1 1	1 1 1	1 1

The effects of all types of faults on the proposed implementations have been evaluated and the percentage of resulting errors are presented in Table III. Results of this table have been obtained from evaluations at three phases. In the first phase, we assumed inputs of the voters are fault-free, and a fault occurs inside of the voter. This fault results in missing or repeating one of the gates of the voter. In the second phase we assumed the voter is fault-free, but a fault in circuit resulted in a bit-flip on one of the inputs of the voter. In the third phase, we assume a fault has occurred inside the voter while another fault in the circuit results in a bit-flip on one of the voter inputs.

IV. DIAGNOSIS SCHEME

The proposed diagnosis scheme is a hierarchical scheme which provides different diagnosis resolution requiring different diagnosis steps. This scheme provides online diagnosis of permanent and transient faults, during runtime operation of circuit. The diagnosis resolution can vary from a *stage* of a TMR block (the sub-circuit between the two consecutive voters), to multiple blocks of TMR stages in distributed TMR architectures.

Table III
FAULT ANALYSIS FOR MTV AND RTV

	MTV			RTV		
	Maskable	Recoverable	Unrecoverable	Maskable	Recoverable	Unrecoverable
(1) Single Missing/Repeated Gate	50%	50%	-	87.5%	12.5%	-
(2) Single Bit-flip on inputs	100%	-	-	100%	-	-
Both (1) and (2)	34.4%	50%	15.6%	56.2%	42.1%	1.7%

A. Single-Block Level Diagnosis

While MTV, as can be seen in Table I, does not provide any diagnosis information, however RTV provides useful information on D_1 and D_2 which can be used to indicate faulty input line. Without loss of generality, here we consider bit-flips on inputs of the voter. This bit-flip may be due to a single fault in functional block (MGF or RGF), or multiple faults on the voter of previous TMR stage. The faulty input (indicating the block producing faulty values) is determined by garbage outputs of RTV. Table IV shows diagnosis information of a TMR block by using garbage outputs resulting from the voter of that stage (D_1 and D_2). The resolution of diagnosis depends on the granularity of the triplicate sub-circuits. In other words, if each TMR stage consists of three copies of a sub-circuit with k gates along with an RTV voter, the resolution of diagnosis is a block of $k+8$ gates (k gates of a copy of functional block and 8 gates of RTV).

Fig. 5 shows an example of a TMR system using an RTV block. Three different values are placed for each node corresponding to three cases: 1) Fault-free situation (left-most value). 2) Fault on DI_2 (middle value): fault is masked in this case and the address of the faulty input (DI_2) is produced on the address outputs of the RTV, i.e. $A_1A_0 = 10$. 3) There are two faults in the circuit (right-most value): one on the DI_1 (data input) and the other one on control input C_1. These faults affect one of the data outputs and the address of faulty data input is indicated by address outputs ($A_1A_0 = 01$ since DI_1 is faulty). Note that garbage outputs only show the location of the fault on data inputs which is used to identify faulty sub-circuits for the repair/replacement phase.

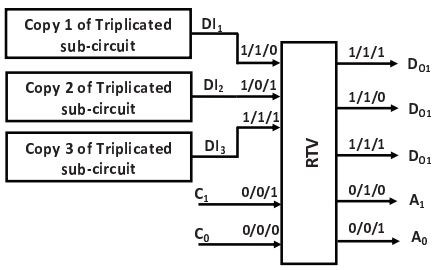

Figure 5. RTV in a TMR system: Address lines and data outputs determine the fault location (values in fault-free case/values in the presence of a single fault on DI_2/values in the presence of two faults on DI_1 and C_1)

B. Multiple-Blocks Level Diagnosis

Indicating the existence/location of a fault and initiating fault diagnosis are done using garbage/diagnosis outputs of the gates. Each RTV produces two diagnosis outputs. Therefore, using n RTVs in a distributed TMR circuit results in $2n$ diagnosis lines. In order to identify the existence/location of the faults these lines are monitored. Observing $2n$ lines iteratively imposes a very high diagnosis time. In order to alleviate this, a reversible circuit is introduced that collects these diagnosis lines to a single output, which indicates the existence of (masked) faults in the circuit. It also keeps the detailed information regarding faulty blocks.

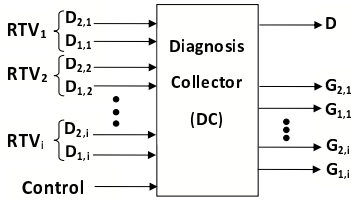

Figure 6. Module collecting the diagnosis information of i RTVs

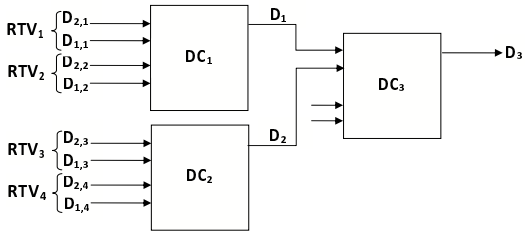

Figure 7. Cascaded DCs to collect diagnosis information of 4 RTVs

Fig. 6 shows the schematic of the circuit which collects diagnosis lines of a set of voters. This module is called *Diagnosis Collector (DC)*. The inputs of this circuit are diagnosis outputs of a set of voters. The function output of this circuit is D which is 1 when at least one of the input lines is 1. The other outputs are copies of the corresponding inputs. By monitoring only the output D, the existence of fault represented by any RTV in the circuit is identified. Thereafter, the diagnosis procedure is initiated to identify individual faulty blocks using the information on other outputs of this gate (actual RTV outputs). This module can also provide a trade off between diagnosis resolution and diagnosis time. Clusters of RTVs can be connected to a cascaded structure of DCs as shown in Fig. 7. This way diagnosis procedure can identify a faulty cluster of RTVs up to the desired resolution considering the required diagnosis time. However, keeping the information at the inputs of a DC module results in more control inputs as well as garbage outputs in DC implementation. An implementation of a DC over 4 RTVs is presented in Fig. 8. As seen in this figure, the implementation consists of a constant garbage output (the output which constantly is 1). This constant output can be used as an input for another DC module. Therefore, the number of garbage outputs does not increase (except one garbage output for all of the DC modules). However,

Table IV
OFFLINE FAULT DIAGNOSIS BASED ON ADDRESS OUTPUTS OF RTV

D_1 D_2	Data values of O_1, O_2, and O_3	Fault location
0 0	$O_1 = O_2 = O_3$	no fault on data inputs / no fault on control inputs
	$O_2 = O_3 \neq O_1$	no fault on data inputs / C_2 is faulty
	$O_1 = O_3 \neq O_2$	no fault on data inputs / C_1 is faulty
	$O_1 = O_2 \neq O_3$	no fault on data inputs / Both C_2 and C_1 are faulty
0 1	$O_1 = O_2 = O_3$	I_1 is faulty / no fault on control inputs
	$O_2 = O_3 \neq O_1$	I_1 is faulty / C_2 is faulty
	$O_1 = O_3 \neq O_2$	I_1 is faulty / C_1 is faulty
	$O_1 = O_2 \neq O_3$	I_1 is faulty / Both C_2 and C_1 are faulty
1 0	$O_1 = O_2 = O_3$	I_2 is faulty / no fault on control inputs
	$O_2 = O_3 \neq O_1$	I_2 is faulty / C_2 is faulty
	$O_1 = O_3 \neq O_2$	I_2 is faulty / C_1 is faulty
	$O_1 = O_2 \neq O_3$	I_2 is faulty / Both C_2 and C_1 are faulty
1 1	$O_1 = O_2 = O_3$	I_3 is faulty / no fault on control inputs
	$O_2 = O_3 \neq O_1$	I_3 is faulty / C_2 is faulty
	$O_1 = O_3 \neq O_2$	I_3 is faulty / C_1 is faulty
	$O_1 = O_2 \neq O_3$	I_3 is faulty / Both C_2 and C_1 are faulty

for each DC module an extra control line is added to the circuit.

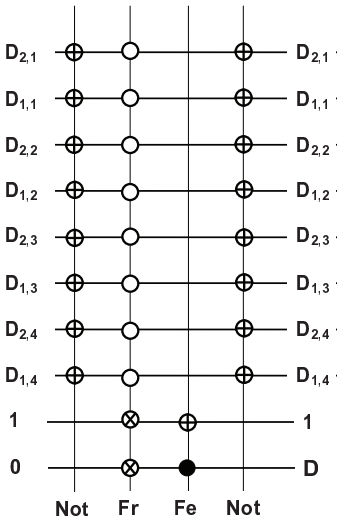

Figure 8. An implementation of a DC for 4 RTVs

V. EXPERIMENTAL RESULTS

In order to evaluate the effectiveness of the proposed voters in terms of fault masking coverage, overhead, and diagnosability, fault simulations on reversible benchmark circuits were carried out. The simulation environment is a program written in C++. This program gets the original reversible circuit as the input and creates the TMR circuit by triplication of individual gates and adding voters to the triplicated circuit. Faults, based on MGF and RGF models, are then injected in random places (individual gates) in the circuit (on both original gates as well as the voters) and the percentage of the cases the faults are not masked is identified through fault simulation.

Variety of algorithms and techniques have been proposed in the literature for logic synthesis of reversible circuits (e.g. [18]). Based on these algorithms, different tools have been developed. However, in this paper RMRLS synthesis tool [19] was utilized in order to prepare reversible benchmark circuits to estimate relative overhead. RMRLS library includes Toffoli, Fredkin, Peres and Reverse Peres basic reversible gates. Synthesized circuits are represented in *netlist* format. As shown in Table V, the five largest benchmark circuits have been chosen from the reversible benchmarks that accompany the RMRLS tool. Moreover, in order to evaluate the proposed technique for larger reversible circuits, SIS synthesis tool was also utilized. The other five benchmark circuits in Table V are created by transforming the synthesized circuit from SIS to reversible counterparts. In this process, circuits are synthesized using basic logic gates, i.e. 2-input AND, NAND, OR, and NOR gates. Then the gates are replaced with their reversible equivalent using basic reversible gates (for example logical OR is implemented by 3 Toffoli gates). This general approach can be applied for reversible synthesis of any logic circuit by performing 1- synthesizing the circuit using basic (traditional) logic gates 2- replacing basic gates with their reversible implementations 3- triplicating all reversible gates 4- adding voters to the output of individual gates.

Table V represents the area of the original circuits as well as the area for the TMR circuits using the proposed MTV and RTV reversible voters. The results suggest that the area overhead of the MTV and RTV implementations are about 17.2% and 37.8% with respect to the area of triplicated circuits. The voter area overhead is calculated as the area of voter over the area of triplicated circuit. Since there is no precise delay model for reversible circuits, their relative delays are compared based on number of stages (logic depth). In this comparison it is assumed that reversible gates have unity delays and also the delay of a circuit is proportional to its logic depth. Therefore, the performance overheads of the voters are shown in terms of reversible circuit depth (stages). The results show that the performance overhead of the circuits obtained directly from reversible benchmarks are lower than that of circuits obtained from SIS synthesized circuits. This supports the motivation for specific synthesis techniques for reversible circuits to reduce parameters such as performance overhead, number of garbage lines, etc. Note that, power consumption characteristics of appropriate technologies for reversible circuits, e.g. quantum computers,

Table V
AREA OVERHEAD OF THE TWO PROPOSED REVERSIBLE VOTERS. # G: NUMBER OF GATES, # IN: TOTAL NUMBER OF INPUTS

		Original		TMR with Voter				Voter Overhead					
				MTV		RTV		MTV			RTV		
Circuit		# G	# In	# G	# In	# G	# In	# G	# In	depth	# G	# In	depth
RMRLS	c.28_2.78	19	38	65	131	75	154	14.0%	14.9%	0	31.6%	35.1%	21%
	hwb5.26	27	250	278	3605	333	3736	14.8%	3.2%	0	42.3%	6.0%	15%
	c.28_2.56	56	858	196	2626	231	2709	16.7%	2.0%	0	37.5%	5.2%	6%
	c.15_2.27	26	76	90	252	105	287	15.4%	10.5%	50%	34.6%	25.9%	130%
	g.c.20.19	78	1175	93	774	108	809	18.8%	2.3%	21%	33.3%	7.9%	39%
SIS	XOR5	82	278	290	908	345	1040	17.9%	8.9%	100%	40.2%	24.7%	180%
	misex1	102	335	362	1100	432	1268	18.3%	9.5%	40%	41.2%	26.2%	75%
	rd53	166	544	590	1790	705	2066	18.5%	9.7%	44%	41.6%	26.6%	92%
	o64	171	732	609	2361	729	2649	18.7%	7.5%	107%	42.1%	20.6%	193%
	clip	1025	3298	3659	10913	4389	12665	19.0%	10.3%	24%	42.7%	28.0%	60%

are fundamentally different from traditional CMOS VLSI technologies. Therefore, power consumption analysis and comparison for reversible circuits, at the current research stage, is meaningless.

In these experiments, there is a voter for every 7 gates in the circuit, to reduce voter area and performance overheads.

Figure 9 presents fault masking coverage of TMR circuits created using MTV and RTV as voters for different number of multiple injected faults. Each data point in this graph is the average coverage among all simulated circuits. In our experiments, the number of fault injections for each data point per circuit was at least 12,500. In these experiments, MGF, RGF, and bit-flip were considered as the reversible fault models. A fixed number of faults are injected into random locations of the circuit and the fault masking coverage for the corresponding number of faults in the circuit is calculated. The coverage for single fault is around 94% for MTV and 100% for RTV, verifying that RTV solves the problem of single point of failure in TMR circuits. Increasing the number of faults decreases fault masking coverage for both voter implementations. However, RTV provides better fault masking coverage than MTV as the number of multiple faults increases. So, a simple and yet reasonable conclusion is that MTV is suitable only for the systems with low error rates when the area is a major concern.

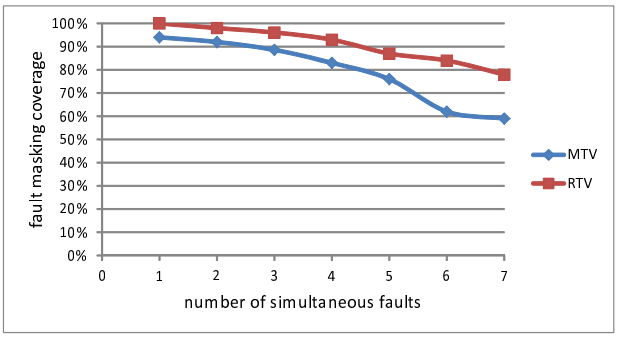

Figure 9. Fault masking coverage for MTV and RTV

VI. SUMMARY AND CONCLUSIONS

Reversible logic has received considerable attention in ultra low power computing and in particular quantum computing. However, due to high failure rates in emerging technologies, fault tolerant design is an important issue for reversible circuit design.

In this paper we proposed different implementations of reversible voters to provide efficient fault masking. The proposed voters provide information about location of the fault for online and offline diagnosis. They can also provide diagnosis information while they mask fault(s) in the circuit. Such diagnosis information can be used in manufacturing test as well as online repair.

REFERENCES

[1] J. Donald and N. K. Jha. Reversible logic synthesis with fredkin and peres gates. *J. Emerg. Technol. Comput. Syst.*, 4(1):1–19, 2008.

[2] A. Barenco, C. H. Bennett, R. Cleve, D. P. DiVincenzo, N. Margolus, P. Shor, T. Sleator, J. Smolin, and H. Weinfurter. Elementary gates for quantum computation. *Phy. Rev. A*, 52(5):3457-3467, 1995.

[3] M. Saeedi, M.S. Zamani, and M. Sedighi. On the behavior of substitution-based reversible circuit synthesis algorithms: Investigation and improvement. *ISVLSI*, 428–436, 2007.

[4] H. M. H. Babu, M. R. Islam, A. R. Chowdhury, and S. M. A. Chowdhury. Reversible logic synthesis for minimization of full-adder circuit. *ESDSD*, 50–54, 2003.

[5] H. M. H. Babu and A. R. Chowdhury. Design of a reversible binary coded decimal adder by using reversible 4-bit parallel adder. *VLSI Design*, 255–260, 2005.

[6] K. N. Patel, J. P. Hayes, and I. L. Markov. Fault testing for reversible circuits. *TCAD*, 23(8):1220–1230, 2004.

[7] K. N. Patel, J. P. Hayes, and I. L. Markov. Fault testing for reversible circuits. *VTS*, 410–416, 2003.

[8] D. P. Vasudevan, P. K. Lala, and J. P. Parkerson. A novel approach for on-line testable reversible logic circuit design. *ATS*, 325–330, 2004.

[9] B. Parhami. Fault-tolerant reversible circuits. *ACSSC*, 1726–1729, 2006.

[10] N. Farazmand, M. Zamani, and M. B. Tahoori. Online fault testing of reversible logic using dual rail coding. *IOLTS*, 204–205 2010.

[11] I. Polian, T. Fiehn, B. Becker, and J. P. Hayes. A family of logical fault models for reversible circuits. *ATS*, 422–427, 2005.

[12] J. P. Hayes, I. Polian, and B. Becker. Testing for missing-gate faults in reversible circuits. *ATS*, 100–105, 2004.

[13] C. H. Bennett. Logical reversibility of computation. *IBM JR&D*, 17(6):525–532, 1973.

[14] Z. Jing and J. C. Muzio. Analyzing fault models for reversible logic circuits. *CEC*, 2422–2427, 2006.

[15] J. A. Abraham. Reliability analysis of digital systems protected by massive redundancy. *Stanford University, PhD Thesis*, 1974.

[16] G. Yang, W. Hung, X. Song, and M. Perkowski. Majority-based reversible logic gates. *Theoretical Computer Science*, 334(1-3):259–274, 2005.

[17] P. O. Boykin and V. P. Roychowdhury. Reversible fault-tolerant logic. *DSN*, 444–453, 2005.

[18] D. Michael Miller, Gerhard W. Dueck, and Dmitri Maslov. A synthesis method for mvl reversible logic. *ISMVL*, 74–80, 2004.

[19] J. Donald and N. K. Jha. Reversible logic synthesis with fredkin and peres gates. *J. Emerg. Technol. Comput. Syst.*, 4(1):1–19, 2008.